Dielectric strength

The maximum point at which the dielectric field withstands the potential difference, without damage is known as di-electric strength.

1.11.4 Functions of a Capacitor

1. It stores electrical energy particularly DC when connected in parallel.
2. It blocks DC supply when connected in series with DC.
3. It couples AC to the next circuit and blocks DC to that next circuit.
4. It discharges while connected to a load.
5. It bypasses unwanted signals to earth.
6. It filters ripples in the pulsating dc to get pure dc while connected in parallel.
7. It decouples the unwanted signals.
8. It helps in producing signals of various frequencies in oscillator circuits.
9. Presently, very high value ultra capacitors are used in place of batteries for quick charging.
10. In a circuit, if an insulator is accidentally placed in between two conductors, it also works as a capacitor. It is called as stray or parasitic capacitance. While designing circuits, the design engineers take care of this fact.
11. Now a days, capacitive type touch screens are widely used.

1.11.5 Capacitive Reactance(X_c)

It is the property of a capacitor which opposes alternating current, is known as Capacitive Reactance. It is represented by the letter X_c. Capacitive reactance is inversely proportional to the frequency applied to the capacitor and the capacitance value of the capacitor. Its unit is also ohm.

$$X_C = \frac{1}{2\pi fc}$$

where f is the frequency and C is the capacitance of the capacitor.

High value capacitor offers low reactance to low frequency, but offers high reactance to high frequency.

Low value capacitor offers low reactance to high frequency signals but offers high reactance to low frequency signals.

Ex: High value capacitors such as 1000 µF, 2200 µF and 4700 µF are used for filtering low frequency 50 Hz ripples in power supplies.

1.11.6 Capacitance

The capacitance (C) is derived from the ratio between charge (Q - Coulomb) and the potential difference between the two plates(V).

$$C = \frac{Q}{V}$$

$$\text{Capacitance (C)} = \frac{\text{Charge (Q)}}{\text{Voltage (V)}}$$

(C in farad, charge in coulombs, V in volts.)

1.11.7 Classification of Capacitors

These are two main classifications

1. Fixed capacitor
2. Variable capacitor

Fixed capacitor



Figure 1.54 (a)

ELEC_EQUIP_Chapter01.indd 27 11/10/2020 12:22:46 PM